Acquired 130.16.32.0/24

Admin = 100
North = 25
East = 20
South = 45

Subnet bits = $32 - \log_2 100$ – Subnet mask
= 32 – 7 – 24
= 1

130.16.32.0 → 130.16.32.0 000 0000 (← host →)
= 130.16.32.0 /25

130.16.32.0 → ~~130.16.32.01 000 00000~~
~~130.16.32.128~~ /25

Assign 130.16.32.0/25 to Admin
its IP range
130.16.32.0 000 0000 = 130.16.32.0
0 111 1111 = 130.16.32.127

for south=45
130.16.32.128/25

Subnet bits = $32 - \log_2 45 - 25$
= 32 – 6 – 25
= 1

/25
130.16.32.1 000 0000 →
→ 130.16.32.1000 0000
= 130.16.32.128/26
= 130.16.32.192/26
→ 130.16.32.1100 0000

Assign 130.16.32.128 /26 to South

Usable IP range =

~~130.16.32.128~~

130.16.32.1000 0000

→ 130.16.32.~~128~~ 1000 0000
= 130.16.32.128

→ 130.16.32.1011 1111
= 130.16.32.191

———— x ———— x ————

for North = 25

130.16.32.192 /26

Subnet bits = 32 - 5 - 26 = 1

SBM = /26

130.16.32.11 00 0000

→ 130.16.32.1100 0000
= 130.16.32.192/27

→ 130.16.32.1110 0000
= 130.16.32.224 / 27

Assign 130.16.32.192 /27 to North

~~usable~~ host range (not all usable)

130.16.32.1100 0000

→ 130.16.32.1100 0000
= 130.16.32.192 / ~~28~~

= 130.16.32.223 / ~~28~~

→ 130.16.32.1101 1111

For East = 20

~~Subnet 1~~ Assign 130.16.32.224/27

Calculating usable host range:

130.16.32.1110 0000

→ 130.16.32.1110 0000
130.16.32.224

→ 130.16.32.1111 1111
= 130.16.32.255

~~130~~.